Rahul Kaushal

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Technical Skills

Languages & Backend: Python, Go, SQL, Bash, FastAPI, asyncio, Pydantic, SQLAlchemy, REST/WebSocket APIs, pytest, Behave (BDD), Locust
Data & ML: PySpark, Databricks, Delta Lake, Spark Structured Streaming, PostgreSQL, PGVector, MLflow, LangChain, LLM/RAG, Azure OpenAI, Vertex AI
Infrastructure: Kubernetes (MicroK8s, AKS), Docker, Ansible, Ceph, Pacemaker/Corosync, Azure (Event Hub, IoT Hub, Blob Storage), GitHub Actions, Prometheus/Grafana

Professional Experience

Advanced Software Engineer — *May 2026 – Present*
Honeywell — *Forge Platform*

- Root-caused a watchdog feedback loop killing idle appliances: a hard MicroK8s kill corrupted Calico's IP state until the pool ran out, and the watchdog's own restart re-triggered it. Fixed with image pre-caching, a Recreate rollout for the single-writer registry volume, and a circuit breaker halting auto-restarts after 5 escalations in 24h.
- Cut incremental-update install time 32.5% (90m11s to 60m51s, 44-package run) after finding a fixed datastore-drain pause, not any install script, was the largest cost — consolidated 13 sequences into 9, halved the pause, moved extraction to `pigz`.
- Built the storage bootstrap letting a node join with or without Ceph healthy: `ceph-csi` as default storage class when Ceph is up, otherwise hostpath with a scale-down/copy/switch/scale-up migration, since recreating a PVC on Ceph loses data.
- Implemented 2-node and N-node Pacemaker/Corosync VIP failover — 90s on graceful standby, 120s from a simulated partition — with a 5-phase test that diffs data on the new primary before passing.
- Built EdgeOrchestrator, the Go service exposing the appliance's REST/WebSocket control-plane API, driving Ansible through argument-array exec rather than a shell, and tracking its own replacement across self-update so a crashed update is not reported as success.

Go, Kubernetes (MicroK8s), Ansible, Ceph, Pacemaker/Corosync, Azure IoT Hub, TypeScript, Prometheus/Grafana

Software Engineer II — *Jul 2024 – Apr 2026*
Honeywell — *ForgeML*

- Built the deployment path routing models to Databricks Serving, AKS or edge devices from one API, with discriminated-union request types validating each target's config at the boundary and status tracked by two independent asyncio tasks.
- Built the asyncio Event Hub processor: three parsers behind a Factory Pattern, concurrent PostgreSQL and Unity Catalog writes, and Blob checkpoints committed only after a successful write — at-least-once delivery with upserts, so a replay overwrites rather than duplicates.
- Routed model registrations through five security scanners (Coverity, Black Duck, Twistlock, ModelScan, AST parser) behind one Strategy Pattern interface, so adding a scanner meant a new class rather than another conditional.
- Fine-tuned Gemini on Vertex AI to translate SkySpark Axon rules into Honeywell's Forge DSL, with a versioned prompt store and tuning-job state machine. A translation counted as correct only if it produced the same results as the original rule, checked by the domain experts who supplied the training pairs.
- Built the SkySpark migration service (SCRAM auth, Zinc/Trio parsing, parallel writes roughly halving write time) and the Behave acceptance suite: 50+ scenarios across six workflow domains run in parallel per tenant, plus Locust load tests.

Python, FastAPI, Databricks, Azure Kubernetes (AKS), Google Vertex AI, MLflow, Azure Event Hub, PostgreSQL

Data Engineer II — *Jun 2023 – Jun 2024*
Honeywell — *Forge Digital Services*

- Built 15+ PySpark ingestion pipelines on Databricks across 12+ source systems (SQL Server, Salesforce, SharePoint, field-service databases) into Delta Lake, using timestamp watermarking for incremental loads.
- Engineered a Spark Structured Streaming pipeline over PDF arrivals in Blob Storage with SCD Type 2 upserts, reconciliation against processed records to catch source deletions, and Change Data Feed so the vectorization job read only changed rows.
- Built the Maintenance Assist chatbot's agent executor — tool selection, chat history, summarization — YAML-configured so each tenant ran its own LLM and prompts without a code change, plus async guardrails alongside the call and a PGVector memory store retrieving turns by max inner product similarity.
- Wrote the SQL transformation pipelines behind field service analytics — incident resolution trends, contract and maintenance metrics, NPS — feeding the customer-facing DSP dashboard.

PySpark, Databricks, Delta Lake, PostgreSQL, LangChain, PGVector, Azure OpenAI, MLflow

Open Source

aio-libs/async-lru | *merged* — *Mar 2026*
Added `cache_contains()` to `alru_cache`: a side-effect-free key lookup that does not move hit/miss counters or reorder the LRU, which the public API had no way to do. Shipped with its own test module and README entry.

facebookresearch/exca | *merged* — *Mar 2026*
Five fixes across six files in Meta's experiment-caching library: `__eq__` comparing the wrong fields, eviction keeping the wrong entries, cleanup leaving files behind, missing f-string prefixes, and a branch raising `UnboundLocalError`.

DepKeeper | *Python, Click, httpx, asyncio* — *2025 – Present*
A CLI that upgrades `requirements.txt` without crossing a major version boundary, preserving your upper bounds and exclusions. Deliberately not a resolver and not a lock file. On PyPI under Apache 2.0.

Education

B.Tech Biotechnology + M.Tech Data Science — *2018 – 2023*
Indian Institute of Technology Madras — *CGPA: 8.77*